

IT Infrastructure and Support Solution

Task 1: Business Requirements Analysis

1.1 Company Overview

Ubuntu Innovations (Pty) Ltd is a technology startup based in Cape Town with 25 employees distributed across multiple departments including Finance, HR, Sales, Management, and Software Development

1.2 Organisational Structure

Ubuntu Innovations currently employs 25 staff members across five departments:

Department	Number of Employees	Primary IT needs
Executive Management	3	Email, file access, reporting tools, VPN, laptops
Finance	4	Accounting software, secure file storage, spreadsheets
Human Resources	3	HR management software, confidential file storage, email
Sales and Marketing	7	CRM, email, presentation tools, internet access
Software Development	8	IDEs, version control (Git), terminal access, local servers
Total	25	

1.3 Business Applications Required

The following applications are required across the organization:

Application	Department(s)	Purpose
Microsoft 365 / Google Workspace	All departments	Email, documents, spreadsheets, presentations
Sage Accounting / Xero	Finance	Invoicing, payroll, financial reporting
Zoho HR / BambooHR	Human Resources	Leave management, employee records, onboarding
Visual Studio Code / IntelliJ IDEA	Software Development	Code editing and development
GitHub / GitLab	Software Development	Version control, code collaboration

Slack / Microsoft Teams	All departments	Internal communication and collaboration
Veeam / Acronis	IT Administration	Backup and disaster recovery

1.4 Data Security Requirements

Ubuntu Innovations handles sensitive business, financial, and personnel data. The following data security requirements have been identified:

- All user accounts must be protected with strong passwords (minimum 12 characters, including uppercase, lowercase, numbers and special characters).
- Multi-factor authentication must be enabled for all user accounts, especially those with administrative or financial access.
- Department folders must be access controlled so that only authorized users can read, write, or modify files.
- All devices must have antivirus software installed and kept up to date.
- Sensitive data belonging to Finance and HR departments must be stored in encrypted folders.
- A regular backup schedule must be implemented to prevent data loss

1.5 Connectivity Needs

- Wired Ethernet connections for all desktop computers and servers using Cat6 cabling for high-speed, reliable connectivity.
- Wireless Wi-Fi 6 access for laptops, mobile devices, and guest access, provided through two dual-band access points placed to cover the entire office.
- A business-grade broadband internet connection with a minimum speed of 100 Mbps to support all 25 users simultaneously.
- A VPN solution for remote access by management and authorized staff working off-site.
- The network must be segmented using VLANs to separate traffic between departments (servers, user devices, printers, and guest Wi-Fi).

Task 2: Hardware Inventory

The following hardware has been recommended to meet the operational needs of Ubuntu Innovations. Each item is justified based on the number of employees, department requirements, and business continuity needs.

Asset	Suggested Specification	Business Justification
Desktop Computers	Intel i5, 16GB RAM	For all office-based staff in Finance, HR, Sales and Development who work at fixed workstations.
Laptops	Intel i7, 16GB RAM	For the 3 Executive Management staff and 2 senior developers who require mobility and remote access.
Router / Firewall	Firewall Router	Provides internet connectivity, network security, firewall protection, and VPN support for all 25 users.
Wireless Access Points	Dual-band Wi-Fi	Provides full office wireless coverage for laptops, mobile devices, and guest Wi-Fi. Two units ensure coverage with no dead zones.
NAS Device	2-bay or 4-bay NAS, 8TB RAID 1	Centralised file storage and daily backups. RAID 1 ensures data redundancy against hard drive failure.

Total Hardware Budget Estimate

Asset	Price (Qty)	Total
Desktop Computers	R 8,500 (20)	R 170,000
Laptops	R 16,000 (5)	R 80,000
Router / Firewall	R 12,000 (1)	R 12,000
NAS Device	R 9,500 (1)	R 9,500
Wireless Access Points	R 5,000 (2)	R 6,500
Estimated Total		R 278,000